Assistant Professor
Engineering Systems and Design Pillar
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# Shaowei Lin

**Date of Revision**: Jan 25, 2017

## Education

2006–2011 **Ph.D. Mathematics**, *University of California, Berkeley*.
Advisor: Bernd Sturmfels
Topic: Algebraic Statistics
Thesis: *Algebraic methods for evaluating integrals in Bayesian statistics*

2002–2005 **B.S. Mathematics**, *Stanford University*.
Honors with Distinction, Minor in Computer Science
Advisor: Yakov Eliashberg
Topic: Symplectic Topology
Thesis: *Invariants on Legendrian knots*

## Career History

2016–Present **Singapore University of Technology and Design**.
Engineering Systems and Design Pillar, Assistant Professor

2015–2016 **Institute for Infocomm Research**.
Sense and Sense-abilities, Deputy Head (Research)

2013–2015 **Institute for Infocomm Research**.
Sense and Sense-abilities, Sense-making Group Leader

2012–2013 **Institute for Infocomm Research**.
Sense and Sense-abilities, Scientist

2011–2012 **University of California, Berkeley**.
Mathematical Challenges in Deep Learning, Postdoctoral Scholar

2005–2006 **Institute for Infocomm Research**.
Digital Wireless, Research Officer

## Research Interests

smart cities, internet-of-things architecture, wireless sensor networks
semantic web, linked data, functional programming, type theory
machine learning, neural networks, deep learning, compressive sensing
statistical learning, algebraic statistics, singular learning
algebraic geometry, computer algebra, singularity theory, tropical geometry

## Grants and Projects

Jan-Jun 2017 **MIT-SUTD Teach-the-Teacher Programme**
TtT Visiting Scholar
*Collaboration between* MIT, SUTD

2016-2017 **Hardware Infrastructure for GPU Accelerated Computing**
Co-Principal Investigator
*SUTD International Design Centre Infrastructure Grant*

2016-2019 **Deep Probability Flow and Functional Web for Artificial Intelligence**
Principal Investigator
*SUTD Start-Up Research Grant*

2014–2017 **Urban Microclimate Multiphysics Integrated Simulation Tool**
UM-MIST $I^2R$ Proposal Lead
*Funded by* L2NIC (MND, NRF)
*Collaboration between* $I^2R$ (S&S), IHPC, HDB

2013–2016 **Heterogeneous Sense-making and Learning Networks**
HELEN Principal Investigator
*Funded by* A*STAR SERC
*Collaboration between* $I^2R$ (S&S), SUTD (Tony Quek)

2014 **Data-Driven Research & Future Computing Paradigms**
DDR-FCP Futurescape 2025 Panel Chairperson
*Steered by* A*STAR SERC

2014 **Semantic Web of Things**
SWoT Futurescape 2025 Panel Member
*Steered by* A*STAR SERC

2012–2014 **Wireless sensor networks for real-time, continuous ambient noise mapping**
WSNoise Analytics Lead
*Funded by* NEA

2012–2014 **Dense, rural, large scale deployment of S&S technologies for vertical farming**
SkySense Analytics Lead
*Funded by* A*STAR SERC
*Collaboration between* $I^2R$ (S&S), Sky Greens

2011–2012 **Mathematical Challenges in Deep Learning**
Postdoctoral Scholar
*Funded by* DARPA, USA
*Collaboration between* Berkeley (Bernd Sturmfels), Stanford (Andrew Ng)

## Honours and Awards

2015 Borderless Silver Award (UM-MIST, for interagency research), MTI

2014 Finalist at World Smart Cities Award (SkySense, for smart agriculture), Spain

2014 Borderless Award (Urban Systems Initiative, for interdisciplinary research), A*STAR

2014 TALENT Award (S&S, for nurturing young scientists), A*STAR

2006 National Science Scholarship (Ph.D.), A*STAR

2005 Roll of Honour, A*STAR

2003–2005 Chairman's Honours List, A*STAR

2005 Mathematics Undergraduate Research Award, Stanford, USA

2004 William Lowell Putnam Mathematical Competition – Top 15 Ind, USA

2002 William Lowell Putnam Mathematical Competition – Top 40 Ind, USA

2002 National Science Scholarship (B.S.), A*STAR

1997–1999 International Mathematical Olympiad, Singapore Team – 3 Bronze Medals

## Professional Activities

### Conference Technical Program

2014 **PHYSCOMNET.** *Workshop on Physics-Inspired Paradigms in Wireless Communications and Networks.*

2014 **IEEE ISSNIP.** *International Conference on Intelligent Sensors, Sensor Networks and Information Processing.*

2014 **IEEE VTS APWCS.** *Vehicular Technology Society Asia Pacific Wireless Communications Symposium*

2014 **DAMLCity.** *Workshop on Data Analytics and Machine Learning for Smart Cities*

### Conference Organization

2015 **IEEE ISSNIP.** *International Conference on Intelligent Sensors, Sensor Networks and Information Processing.*

**RIoT.** *International Conference on Recent Advances in Internet of Things.*

Local Arrangements Coordinator.

2014 **IEEE ISSNIP.** *International Conference on Intelligent Sensors, Sensor Networks and Information Processing.*

Symposium and Tutorial Coordinator.

### Miscellaneous

2016 **SGInnovate.**

Consultant for Artificial Intelligence.

2016 **SUTD Deep Learning Day.**

Co-organizer with NVIDIA.

2016 **Research Workshop on the Future of AI.**

Co-organizer with NVIDIA.

## Invited Talks and Tutorials

2016 Talk: SGInnovate AI Evening, "Intentional intelligence", SGInnovate, Singapore

2016 Talk: SUTD-ZJU IDEA Workshop, "Smarter cities through distributed artificial intelligence", ZJU, Hangzhou

2016 Panellist: Cloud Asia ICCCRI, "Game changers in HPC for artificial intelligence", Suntec, Singapore

2016 Talk: Sixth Singapore Conference on Statistical Science, "Understanding the curse of singularities in machine learning", NUS, Singapore

2016 Tutorial: CSD&M Asia, "Big data and data analytics", SUTD, Singapore

2015 Talk: SIAM Conference on Applied Algebraic Geometry, "Lessons on statistical singularities from deep learning", Daejeon, South Korea

2014 Tutorial: IEEE ISSNIP, "Sensing and sense-making architectures for the urban superorganism", Singapore

2014 Talk: NIMS Workshop on Algebraic Statistics, "Deep learning from cat videos to algebraic singularities", Daejeon, South Korea

2013 Tutorial: CMND Thematic Program on Motivic Invariants and Singularities, "Singular learning theory", Undergraduate Summer School, Notre Dame, USA

2012 Talk: Workshop on Algorithms for Modern Massive Data Sets (MMDS), "Studying model asymptotics with singular learning theory", Stanford, USA

2012 Talk: AIM Workshop on Singular Learning Theory, "Singular Learning Theory: a view from Algebraic Geometry", Palo Alto, USA

## Publications

### Journal Papers and Preprints


[1] C. J. Hillar, S. Lin, and A. Wibisono. "Inverses of symmetric, diagonally dominant positive matrices and applications". *submitted to SIAM Journal on Matrix Analysis and Applications, arXiv:1203.6812* (2017).

[2] Z. Liu, S. Lin, T. Q. S. Quek, and W. Zhang. "Robustifying Deep Heterogeneous Sensor Data Fusion". *submitted to Journal of Select Topics in Signal Processing* (2016).

[3] S. Lin. "Asymptotic approximation of marginal likelihood integrals". *accepted in Journal of Algebraic Statistics, arXiv:1003.5338* (2016).

[4] M. Drton, S. Lin, L. Weihs, and P. Zwiernik. "Marginal likelihood and model selection for Gaussian latent tree and forest models". *to appear in Bernoulli* (2016).

[5] M. A. Alsheikh, D. Niyato, S. Lin, H.-P. Tan, and D. I. Kim. "Fast adaptation of activity sensing policies in mobile devices". *accepted in IEEE Transactions on Vehicular Technology* (2016).

[6] M. A. Alsheikh, S. Lin, D. Niyato, and H.-P. Tan. "Rate-Distortion Balanced Data Compression for Wireless Sensor Networks". *Sensors Journal, IEEE* 16.12 (2016), pp. 5072–5083.

[7] M. A. Alsheikh, D. Niyato, S. Lin, H.-P. Tan, and Z. Han. "Mobile big data analytics using deep learning and Apache Spark". *Network, IEEE* 30.3 (2016), pp. 22–29.

[8] M. A. Alsheikh, D. T. Hoang, D. Niyato, H.-P. Tan, and S. Lin. "Markov Decision Processes with Applications in Wireless Sensor Networks: A Survey". *Communications Surveys Tutorials, IEEE* 17.3 (2015), pp. 1239–1267.

[9] M. Abu Alsheikh, S. Lin, D. Niyato, and H.-P. Tan. "Machine Learning in Wireless Sensor Networks: Algorithms, Strategies, and Applications". *Communications Surveys Tutorials, IEEE* 16.4 (2014), pp. 1996–2018.

[10] S. Lin, C. Uhler, B. Sturmfels, and P. Bühlmann. "Hypersurfaces and their singularities in partial correlation testing". *Foundations of Computational Mathematics* 14.5 (2014), pp. 1079–1116.

[11] V. I. Morgenshtern, E. Riegler, W. Yang, G. Durisi, S. Lin, B. Sturmfels, and H. Bolcskei. "Capacity Pre-Log of Noncoherent SIMO Channels Via Hironaka's Theorem". *Information Theory, IEEE Transactions on* 59.7 (2013), pp. 4213–4229.

[12] M. A. Cueto and S. Lin. "Tropical secant graphs of monomial curves". *Beiträge zur Algebra und Geometrie* 54.1 (2013), pp. 383–418.

[13] S. Lin. "Algebraic methods for evaluating integrals in Bayesian statistics". PhD thesis. University of California, Berkeley, 2011.

[14] S. Lin and B. Sturmfels. "Polynomial relations among principal minors of a $4\times 4$-matrix". *Journal of Algebra* 322.11 (2009), pp. 4121–4131.

[15] S. Lin, B. Sturmfels, and Z. Xu. "Marginal likelihood integrals for mixtures of independence models". *The Journal of Machine Learning Research* 10 (2009), pp. 1611–1631.

[16] S. Lin, W. W. Ho, and Y.-C. Liang. "Block diagonal geometric mean decomposition (BD-GMD) for MIMO broadcast channels". *Wireless Communications, IEEE Transactions on* 7.7 (2008), pp. 2778–2789.

### Conference Papers

[17] Z. Liu, S. Lin, T. Q. S. Quek, and W. Zhang. "Deep fusion of heterogeneous sensor data". *accepted in International Conference on Acoustics, Speech, and Signal Processing*. 2016.

[18] M. A. Alsheikh, A. Selim, D. Niyato, L. Doyle, S. Lin, and H.-P. Tan. "Deep activity recognition models with triaxial accelerometers". *Workshops at the Thirtieth AAAI Conference on Artificial Intelligence*. 2016.

[19] M. A. Alsheikh, S. Lin, H.-P. Tan, and D. Niyato. "Toward a robust sparse data representation for wireless sensor networks". *Proceedings of IEEE Conference on Local Computer Networks*. 2015.

[20] L. Z. Wong, H. Chen, S. Lin, and D. C. Chen. "Imputing missing values in sensor networks using sparse data representations". *Proceedings of the 17th ACM international conference on Modeling, analysis and simulation of wireless and mobile systems*. 2014, pp. 227–230.

[21] M. Abu Alsheikh, P. K. Poh, S. Lin, H.-P. Tan, and D. Niyato. "Efficient data compression with error bound guarantee in wireless sensor networks". *Proceedings of the 17th ACM international conference on Modeling, analysis and simulation of wireless and mobile systems*. 2014, pp. 307–311.

[22] P. Zhang, J. Y. Koh, S. Lin, and I. Nevat. "Distributed event detection under Byzantine attack in wireless sensor networks". *Intelligent Sensors, Sensor Networks and Information Processing, IEEE 9th International Conference on*. 2014, pp. 1–6.

[23] G. Peters, I. Nevat, S. Lin, and T. Matsui. "Modelling threshold exceedence levels for spatial stochastic processes observed by sensor networks". *Intelligent Sensors, Sensor Networks and Information Processing, IEEE Ninth International Conference on*. 2014, pp. 1–7.

[24] M. A. Alsheikh, D. Niyato, S. Lin, and H.-P. Tan. "Area coverage under low sensor density". *Proceedings of the 11th IEEE Communications Society Conference on Sensor and Ad Hoc Communications and Networks*. 2014.

[25] E. Riegler, V. I. Morgenshtern, G. Durisi, S. Lin, B. Sturmfels, and H. Bolcskei. "Noncoherent SIMO pre-log via resolution of singularities". *Information Theory Proceedings, IEEE International Symposium on*. 2011, pp. 2020–2024.

[26] M. A. Cueto and S. Lin. "Tropical secant graphs of monomial curves". *FPSAC 2010, DMTCS Proceedings AN*. 2010, pp. 669–680.

[27] S. Lin, W. W. Ho, and Y.-C. Liang. "MIMO broadcast communications using block-diagonal uniform channel decomposition (BD-UCD)". *Personal, Indoor and Mobile Radio Communications, IEEE 17th International Symposium on*. 2006, pp. 1–5.

[28] S. Lin, W. W. Ho, and Y.-C. Liang. "Block-diagonal geometric mean decomposition (BD-GMD) for multiuser MIMO broadcast channels". *Personal, Indoor and Mobile Radio Communications, IEEE 17th International Symposium on*. 2006, pp. 1–5.

## Teaching

2016 **SUTD** *Machine Learning (Undergraduate)*
Course Co-Instructor.

2016 **SUTD** *Statistics (Graduate)*
Course Instructor.

2013 **CMND** *Thematic Program on Motivic Invariants and Singularities.*
Undergraduate Summer School (21-25 May) Mini-course on Singular Learning.

2009 **UC Berkeley** *Math 1B Calculus*
Graduate Student Instructor.

2002, 2006 **National University of Singapore** *International Mathematical Olympiad*
Trainer for Singapore National Team.

2002 **Dunman High School** *Upper and Lower Secondary Mathematics*
Relief Teacher for the Gifted Education Programme.

## Mentored Students

2017–Present **Ng Aik Beng** (Ph.D. student)
Project: Artificial Intelligence for Personal Assistants

2016–Present **Lai Zhangsheng** (Ph.D. student)
Project: Machine Reasoning

2016–Present **Gary Phua** (Ph.D. student)
Project: Statistical Physics and Neural Networks

2015–Present **Liu Zuozhu** (Ph.D. student)
Co-advisor: Tony Quek (SUTD)
Project: Distributed Deep Reinforcement Learning

2014–2015 **Steven Dinh Thinh Quang** (Ph.D. student)
Co-advisor: Tony Quek (SUTD)
Project: Multimodal Data Fusion for Sensor Networks

2013–2014 **Poh Puay Kai** (UROP student)
Co-advisor: Seth Gilbert (NUS)
Project: Sensor Data Compression using Neural Networks

2013–2014 **Zhang Wenyu** (Research Attachment)
Project: Deep Learning for Multimodal Sensor Fusion

2013–2014 **Wong Liang Ze** (Research Attachment)
Project: Missing Data Imputation for Sensor Networks

2013–Present **Mohammad Abu Alsheikh** (Ph.D. student)
Co-advisors: Hwee-Pink Tan ($I^2R$), Dusit Niyato (NTU)
Project: Machine Learning Principles for Designing Sensor Networks

2011–2012 **Luca Weihs** (Undergraduate)
Main advisor: Bernd Sturmfels (UCB)
Project: Tensor Tree Cumulants and the Salmon Conjecture

2011–2012 **Andrew Critch** (Ph.D. student)
Main advisor: Bernd Sturmfels (UCB)
Project: Algebraic Statistics